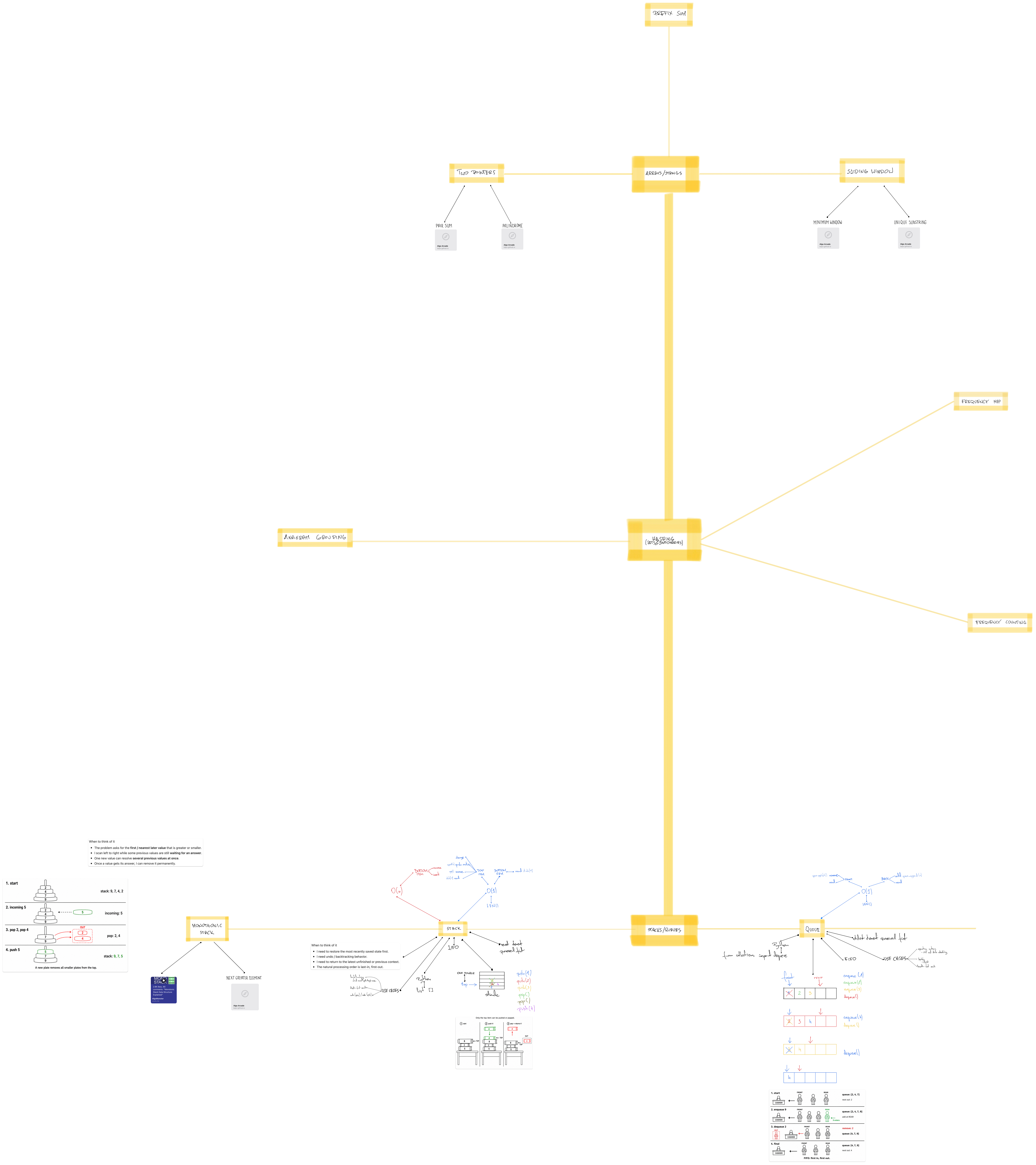
PREFIX SUM
TWO POINTERS
ARRAYS/STRINGS
SLIDING WINDOW
PAIR SUM
Algo Arcade
bajor.github.io
PALINDROME
Algo Arcade
bajor.github.io
MINIMUM WINDOW
Algo Arcade
bajor.github.io
UNIQUE SUBSTRING
Algo Arcade
bajor.github.io
FREQUENCY MAP
ANAGRAM GROUPING
HASHING
(SETS & DICTIONARIES)
FREQUENCY COUNTING
1. start
2
4
7
9
stack: 9, 7, 4, 2
2. incoming 5
5
incoming: 5
3. pop 2, pop 4
OUT
pop: 2, 4
4. push 5
stack: 9, 7, 5
A new plate removes all smaller plates from the top.
MONOTHONIC STACK
MONO STACK
2.9K likes, 90 comments. "Monotonic Stack Data Structure Explained"
AlgoMonster
youtu.be
NEXT GREATER ELEMENT
Algo Arcade
bajor.github.io
BOTTOM ITEM
remove
insert
O(n)
change
append(x) push onto
pop() remove
stack[-1] read
TOP ITEM
BOTTOM ITEM
read stack[0]
O(1)
LEN()
STACK
LIFO
newest element processed first
backtracking
find correct path through maze
depth-first search
undo (pop) / redo (push)
USE CASES
Python
list []
ONE POINTER
top
stack
push(1)
push(2)
push(3)
pop()
pop()
push(4)
Only the top item can be pushed or popped.
① start
② push 4
③ pop -> returns 4
TOP
OUT
8
2
5
4
STACKS/QUEUES
queue.popleft() remove
read
FRONT
BACK
add queue.append(x)
read
O(1)
LEN()
QUEUE
oldest element processed first
Python
from collections import deque
FIFO
USE CASES
operating systems
- CPU and disk scheduling
Spotify
- playlist
breadth-first search
front
rear
1
2
3
enqueue(1)
enqueue(2)
enqueue(3)
dequeue()
2
3
4
enqueue(4)
dequeue()
4
dequeue()
4
1. start
FRONT
REAR
CASHIER
queue: [2, 4, 7]
next out: 2
2. enqueue 9
9 enters
queue: [2, 4, 7, 9]
add at REAR
3. dequeue 2
OUT
remove: 2
queue: [4, 7, 9]
4. final
queue: [4, 7, 9]
next out: 4
FIFO: first in, first out.

When to think of it

- The problem asks for the **first / nearest later value** that is greater or smaller.
- I scan left to right while some previous values are still **waiting for an answer**.
- One new value can resolve **several previous values at once**.
- Once a value gets its answer, I can remove it permanently.

When to think of it

- I need to restore the most recently saved state first.
- I need undo / backtracking behavior.
- I need to return to the latest unfinished or previous context.
- The natural processing order is last-in, first-out.